PA194Q: a different 194-state PA consistency machine

Corpus comparison, a changed axiom emitter, and a second construction with an invariant certificate

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Abstract

We construct **PA194Q**, a deterministic binary single-tape machine with 194 working states, whose blank-tape nonhalting is equivalent to Con(PA) by an ordinary mathematical proof formalizable in PA. **No machine with fewer than 194 working states was established.** The result meets the requested alternative of a materially different 194-state design.

PA194Q exchanges the two premises of the quantifier-distribution axiom emitted by the existing separate-head source. The source transformation was proposed in earlier larger-machine research; the new contribution is its complete 194-state realization, decoder and compatible folds, exact source binding, and independent literal-tape verification. Its resolved source differs at 79 of 3,649 instructions from PA194C. Both proof calculi have the same theorems by explicit, conclusion-preserving proof translations. Its allocation is $134 + 67 - 7 = 194$.

The fresh comparison is pinned to ProveIt commit `17f92105e2bc4317feae6c219cb2143437b482ff`. It includes 41 delivered packets and recognizes 4,417 literal-table occurrences, 620 byte forms and 446 action-graph classes. The eight existing 194-state graphs include a rejected scanner, repeated copies of PA194C, and previously supplied symmetric, alternative-decoder and zero-jump variants. PA194Q is absent even under arbitrary control renaming, change of start, tape reflection and symbol complementation.

All 4,749 arbitrary-register source outcomes pass two physical checks; the independent checker rederives four list lemmas from literal rows. An exact 57,922-step blank initialization reaches the intended complete source frame. A second new 194-state machine, PA194CSZ, combines symmetric Shift with a zero-register jump using a 263,180-pair inductive certificate. The archive contains both complete tables, deterministic builders, source maps, verifiers, certificates, comparison data, this PDF and its complete LaTeX source. The logical PA calibration is explicitly inherited; no new end-to-end Lean or Rocq theorem or completed iPRA endpoint is asserted.

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1 Result and precise machine convention

The principal machine is `machine/pa194q.tm`. Its working states are $0, \ldots, 193$, initial state is 0, and terminal halt label is 194. The single tape is two-way infinite, with alphabet $\{0, 1\}$ and all initial cells zero. The head initially scans cell zero. Each transition writes one bit and moves exactly one cell left or right. The table has 388 rows, two for every working state and none for the halt label. All working states are reachable in the pointed transition graph; this is not a claim that the actual blank computation visits each state.

Each row has the form

```
state  read  write  L-or-R  next-state.
```

The authoritative dense-table SHA-256 is

`48e98b5504dc5b3f20a2d040cb54ed1c96bee4c5e32ea27ae98ffada4dce7ea1`.

The optional second table is `secondary/pa194csz/machine_csz/pa194CSZ.tm`, with the same tape, state, start and halt conventions. Its SHA-256 is

`62e93098caf7c09ff2938d07064f0dbbb47c71ec369892968258fff708592269`.

Appendices A and B print every transition.

Let H_M be the usual arithmetic sentence asserting a finite literal halting trace from the specified blank start. The principal conclusion is

$$\mathrm{PA} \vdash (\neg H_{\mathrm{PA194Q}} \leftrightarrow \mathrm{Con}(\mathrm{PA})).$$

The new preservation theorem is $\mathrm{PA} \vdash H_{\mathrm{PA194Q}} \leftrightarrow H_{\mathrm{PA194C}}$. It is proved through interderivable numerical calculi and their source simulations, not by asserting that the two instruction graphs are equal. The second construction instead has a step-for-step source preservation proof on an inductive invariant. Section 5 separates these new results from the inherited PA calibration.

2 Comparison of the supplied designs

2.1 Scope, identity and evidence

The current repository tree has 36 filed historical PA research packets and five additional incoming-only deliveries. Twenty-one retained incoming ZIPs duplicate filed packets. This gives a 41-delivery catalogue, rather than counting each copied archive as new research. The complete archive-by-archive ledger is in Appendix C and `comparison/archive_catalogue_current.json`.

The fresh literal-table scanner reads every `.tm` and `.txt` in the current PA tree, all current incoming ZIPs and their recursively encountered ZIPs. It recognizes complete numeric five-column binary tables with exactly one terminal target. The same two text extensions are used inside archives. It does not extract arbitrary machine encodings embedded only in Python, Lean, JSON or prose. All frontier machines discussed here have literal tables in scope. No blob or archive read failed. All 606 byte forms from the older census are present; 14 new byte forms have been added.

Quantity	Fresh result
Literal-table occurrences	4,417
Distinct byte forms	620
Action-graph classes	446
194-state byte forms / graph classes	15 / 8
195-state byte forms	26
196-state byte forms	163
208-state byte forms	177
Recursively encountered distinct ZIP payloads	44

These counts include reference copies, parents, synthetic controls and rejected experiments. They are not counts of valid consistency theorems. Two disconnected synthetic controls have no spanning root and are excluded from the graph-class count. The recognized corpus has no 190–193-state table; this is a census observation, not a lower bound.

The canonical graph comparison retains read-bit labels, write bits, movement directions and the halt target, while minimizing the ordered breadth-first serialization over every root reaching the whole working graph. Additional checks allow global tape reflection and simultaneous read/write symbol complementation. A changed start or finite-control renaming therefore cannot explain the reported novelty.

2.2 All eight existing 194-state graphs

Existing design	Distinction and supplied status
PA194-67	Primary separate-head source; $134 + 67 - 7$; 3,673 sites and 4,780 outcomes.
PA194-67-AD	The same logical source with physical A/D register locations exchanged; a different action graph.
PA194-68	Alternative decoder association, $134 + 68 - 8$; same 3,673-site source.
Rejected scanner194	Eight local folds pass, but a legal seven-zero address is mistaken for a separator. Not a valid certified source simulation.
PA194C	Cached-head Shift and MP-to-cache controller; $134+67-7$; 3,649 sites and 4,749 outcomes. Identical principal table in all five newer packets.
PA194CS	Symmetric destructive equality in Shift removes three sites. One added reader is recovered by folding $41 \mapsto 330$; 3,646 sites.
PA194altC	Cached controller with the alternative 68-reader/eight-fold decoder; already an audited search input.
PA194CZ	Cached controller with an invariant-guarded zero-register realization of source jump 26; auxiliary seed with the earlier physical audit.

Thus neither PA194C, PA194CS nor the cached 68/eight-fold composition would be a new answer to this request. PA194Q changes a generated numerical axiom; PA194CSZ is a new compatible combination of two supplied changes. Their evidence is supplied separately from the finite novelty test.

2.3 The 195-, 196- and 208-state families

The important 195-state constructions are the original bundled-diagonal PA195, the separate-head 195S, cached 195C, a distinct intermediate separate 195 layout, the bundled-diagonal cached PA195DC, and M195S8 with its longer return scanner. Their source lengths are respectively 3,700, 3,673, 3,649,

3,673, 3,676 and 3,700 sites. The two successful 194 allocations both leave 60 effective decoder controls: either $67 - 7$ or $68 - 8$. Their savings cannot be added just because their fold counts differ.

The 196-state work includes decoder reassociation, altered helper returns, physical register permutations, cached-head commands, zero-register jumps, and the diagonal and separate-head logical reductions. The NEG18 source saves many primitive instructions yet retains 197 machine states in its supplied implementation. A smaller source DAG or fewer source instructions therefore need not reduce the binary machine: reader sharing, scanner limitations and compatible helper reuse determine the final count.

Several early 208-state reports give conditional-context, entry-context, history and three-cell explanations of the same physical identification. They are alternative presentations of a shared saving, rather than four independent savings. Later helper layouts add distinct constructions, but the much larger raw-table census also contains unsuccessful experiments. The imported James Chang PA209 lineage and the U209 parents are distinct; shared ancestry does not identify every machine with the same state count. The detailed ledger preserves each packet's proposed improvement and reported failure or evidence boundary.

2.4 Formalization status

The current PA README explicitly states that the machine-specific iPRA consistency reductions remain open for the primary PA194, original PA195 and PA196 targets [1]. It distinguishes ordinary finite-run Lean theorems, Python certificates, conditional arithmetic endpoints and a completed weak-arithmetic consistency equivalence. The imported PA196 formal development supplies the PA implication from nonhalting to consistency. The converse cannot be inferred merely from external nonhalting. The supplied later manuscripts give an ordinary syntactic PA calibration addressing that converse; this report uses it as an identified mathematical dependency, and does not relabel it as a new kernel-checked theorem.

3 The new emitter and its 194-state realization

3.1 A changed axiom at unchanged stack position

With $I(a, b) = 4p(a, b)$ and $U(D, A) = 4p(A, D) + 2$, define

$$P = U(D, A), \quad Q = U(D, I(A, H)), \quad R = U(D, H).$$

The existing sixth head is $P \to (Q \to R)$; PA194Q emits $Q \to (P \to R)$. It remains the second generated stack entry below the separate diagonal head. Every other emitted head, guard, command and scheduler component is retained.

This is a different numerical computation. At $A = 1$, $D = H = 0$, $(P, Q, R) = (6, 42, 2)$. The old head has code 31,609,172; the new head has code 76,268. The emitted theorem families are interderivable, but the machine does not enumerate identical codes in identical stores. Exactly 79 of the 3,649 resolved source instructions change. All others, including their targets, agree with PA194C. The complete 190-token AX word differs only in its 19-token sixth-head block. The new source's longest address has length 36, compared with 37 for PA194C.

The premise exchange itself is not claimed as a new logical discovery. It appears among the supplied earlier source-variant proposals, including the PA195 diagonal-combined research [4]. What was not in the pinned corpus is this 194-state literal realization and its full source and physical certificate.

3.2 Allocation and compatible folds

The deterministic compiler first constructs 134 fixed runtime/controller controls and 67 axiom-reader controls. Seven simultaneous contextual folds then give

$$134 + 67 - 7 = 194.$$

The raw table, complete grammar, fold specification and dense/native conjugacy are delivered. The native start and halt are 74 and 254; renumbering by breadth-first traversal gives the published dense table. The relevant folds are:

Removed	Retained	New read-0 action	New read-1 action
2004	2052	`0L2015`	`1L2048`
2021	2030	`0R90`	`1L2023`
2025	2035	`0L2002`	`1R61`
2027	2046	`0L2036`	`1R2031`
2039	2044	`0R2040`	`1L2014`
2055	2056	`0L2042`	`1R2043`
2058	2017	`0L2050`	`1L2062`

For each pair, the preceding tape cell records which old reader was entered. The checker examines both possible current symbols for both entry histories against the *final simultaneous quotient*. All 28 cases preserve the complete exterior tape, end with the intended written bit, head position and projected target, and take one to five physical steps. Checking all folds against the final table is essential: individually correct folds need not be compatible after changing a shared helper.

The incoming-row condition is checked for every affected reader, and all unaffected transitions commute with the control projection. Induction on finite runs gives positive finite-trace preservation on the specified entry invariant. The physical source audit below is stronger than a successful local fold screen: it binds the complete literal table to every source operation and catches return-scanner failures that local fold checks can miss.

4 Physical verification and its logical scope

The logical registers are $(N, T, W, A, D, H, X, Y, L)$, represented by nine unary blocks in a fixed physical order. A source frame contains a binary source address p, a zero gap of length $48 + n - |p|$ with $n \geq 0$, a delimiter, these nine register blocks and an arbitrary finite nonempty spare list. Thus the address-plus-gap length is $48 + n$; no shorter-header contract is claimed. The checks quantify over arbitrary natural register values, both decrement outcomes and arbitrary spare values and length. They do not sample a bounded range of register values. Successor frames stay in this domain: the header parameter never decreases, and an increment appends a one-valued spare while other primitive steps retain the spare list.

The first checker uses the inherited arithmetic-block contracts and verifies all 4,749 primitive outcomes. It takes 1,540,295 symbolic transitions, with at most 497 per outcome. The second checker imports neither the optimizing compiler nor the first checker's arithmetic-block oracle. It rederives four generic list-sweep induction lemmas from the literal rows, resolves the source graph using two independent methods, and then checks all 4,749 outcomes. Its replay uses 1,124,844 symbolic transitions, at most 403 per outcome. Every certified physical duration has a positive constant term, at least three.

Check	PA194Q	PA194CSZ
Working states / rows	194 / 388	194 / 388
Primitive source sites	3,649	3,646
Universal primitive outcomes	4,749	4,746
Independent symbolic transitions	1,124,844	1,100,566
Minimum duration constant	3	3
Exact blank startup transitions	57,922	58,014

A separate bytearray simulator checks the entire blank initialization. PA194Q reaches native state 84 with head at -2, source site 25, logical registers $(2, 0, 0, 0, 0, 0, 0, 0, 0)$ and 26 one-valued spare registers after exactly 57,922 steps. Every cell is compared with the intended complete frame and no earlier halt occurs. This finite startup check establishes initialization; it is not evidence from a long run for nonhalting.

The ordinary proof turning these contracts into a halting equivalence requires positive duration and absence of an intermediate premature halt. Forward source simulation concatenates the certified physical segments. For the reverse direction, a first literal halt at time t is incompatible with $t + 1$ nonhalting source steps: their positive simulations would force one deterministic literal execution to continue nonhalting beyond t. This bounded argument is formalizable in PA, including its nonstandard models. It does not assume the desired nonhalting conclusion.

The executable verifiers are supporting mathematical certificates and trusted programs, not a new foundational proof kernel. The retained Lean files identify inherited interfaces and the existing implicational exchange theorem. This archive does not claim that a new end-to-end Lean or Rocq build proves the consistency biconditional.

5 Logical calibration of PA194Q

The new machine changes one axiom head and retains the cached-head commands of the pinned PA194C construction. Its source instruction map is therefore not asserted to equal PA194C's instruction map. The preservation argument passes through finite derivations in two interderivable numerical calculi, then through their complete command compilers and literal tape simulations.

5.1 The exact change of axiom

Put

$$p(a, b) = \frac{(a + b)(a + b + 1)}{2} + b, \qquad I(a, b) = 4p(a, b), \qquad U(i, B) = 4p(B, i) + 2.$$

Write $a \to b$ for the numerical constructor $I(a, b)$. In the separate-diagonal-head calculus let

$$P = U(D, A), \qquad Q = U(D, I(A, H)), \qquad R = U(D, H).$$

The old sixth head and its new replacement are respectively

$$L_6(D, A, H) = P \to (Q \to R), \qquad L_6^{\mathrm{Q}}(D, A, H) = Q \to (P \to R). \tag{Q1}$$

Every other head, the guarded separation gate, the separate diagonal head, the numerical constructors, and the inference rules are retained. There is no change to the numerical parameters D, A, H within any formula. In particular, (Q1) is an exchange of two premises within one axiom, not a permutation of the three parameter names or an exchange of stack positions.

Both postfix emitters have 19 tokens. Writing A, D, H for the macros that push the old top and load that register, C for the pairing macro, and i for increment, their words are

$$\begin{aligned} L_6 &: A\,D\,C\,i\,i\;A\,H\,C\,D\,C\,i\,i\;H\,D\,C\,i\,i\;C\,C, \\ L_6^{\mathrm{Q}} &: A\,H\,C\,D\,C\,i\,i\;A\,D\,C\,i\,i\;H\,D\,C\,i\,i\;C\,C. \end{aligned} \tag{Q2}$$

The exact emitter audit identifies these words inside the complete 190-token AX word, checks both displayed formula blocks in a free algebra, and verifies that every surrounding token is unchanged. Each block pushes the incoming top and leaves the older stack intact, so the replacement preserves every other emitted stack entry.

5.2 A proof translation independent of quantifier distribution

The unchanged restricted implicational basis derives, uniformly in all numerical codes a, b, c, the exchange formula

$$(a \to (b \to c)) \to (b \to (a \to c)). \tag{Q3}$$

This is already the theorem `Dominated.exchange` in `PA196/Logic/ImplicationBasis.lean`. Its construction uses only the restricted propositional head, modus ponens, reflexivity of the numerical bound, and the two inequalities $a, b \le I(a, b)$. In particular it does not use either version of the sixth head. It also needs no rule treating code one as falsity.

For clarity, (Q3) is an intuitionistic implicational tautology. From assumptions $f : a \to (b \to c)$, b, and a, two applications of modus ponens give c; three applications of the implicational deduction construction give (Q3). The K and S combinators used by that construction are derived from the same unchanged restricted basis. The optional certificate supplied with this report checks a finite K, S, MP derivation of this exact exchange formula.

Apply (Q3) at $(a, b, c) = (Q, P, R)$ and use the new axiom $Q \to (P \to R)$ as its minor premise. This derives the old axiom $P \to (Q \to R)$. Conversely, (Q3) at (P, Q, R) derives the new axiom from the old one. These are uniform finite derivations for every triple of parameter values D, A, H.

Let $\mathcal{C}$ be the pinned separate-diagonal-head calculus and $\mathcal{C}_{\mathrm{Q}}$ the calculus with (Q1)'s replacement. Transform a $\mathcal{C}$ derivation recursively: retain every unchanged axiom; replace a sixth-head leaf by the new sixth-head axiom followed by the first exchange derivation above; recursively transform the premises at modus ponens and generalization nodes. Every conclusion is preserved literally. The converse transformation uses the other exchange derivation. Thus there are primitive-recursive proof transformations F, G whose correctness PA proves:

$$\begin{aligned} \mathrm{Prf}_{\mathcal{C}}(e, B) &\Longrightarrow \mathrm{Prf}_{\mathcal{C}_{\mathrm{Q}}}(F(e), B), \\ \mathrm{Prf}_{\mathcal{C}_{\mathrm{Q}}}(e, B) &\Longrightarrow \mathrm{Prf}_{\mathcal{C}}(G(e), B). \end{aligned} \tag{Q4}$$

There is no substitution of arbitrary formulas into a freshness-restricted gate. Only the numerical parameters of the unrestricted implicational exchange template are instantiated. In particular (Q4) proves, inside PA, equivalence of the two existential refutation predicates with conclusion code one.

The code-zero alias. These are numerical derivations, so they include zero among the allowed formula codes. The identity $I(0, 0) = 0$ never violates a modus-ponens step: its code equality remains exactly the equality required by the rule. In the arithmetic interpretation of formulas, $T(B)$ denotes the formula produced by a syntactic translation, not an arithmetic truth predicate. There is a uniform proof-producing construction of the instancewise PA equivalence

$$T(I(a, b)) \leftrightarrow (T(a) \to T(b)),$$

whose exceptional $(a, b) = (0, 0)$ case is the fixed equivalence of truth with truth implying truth. One must not assert literal syntactic identity in that exceptional case. The new head's three displayed subformulas are universal codes and hence are nonzero; more generally the proof translation does not require any nonzero-code assumption.

5.3 The ten-entry cached command interface

Let $[a; w] = p(a, w)$ and let hd, tl be the total Cantor projections. In particular $\mathrm{hd}(0) = \mathrm{tl}(0) = 0$ and $[0; 0] = 0$. The command store is $\langle t, w, d, h\rangle$. The commands retained from PA194C are

$$S\langle t, w, d, h\rangle = \begin{cases} \mathsf{halt}, & h = 1, \\ \langle h, \mathrm{tl}(w), d, \mathrm{hd}(w)\rangle, & h \neq 1, \end{cases}$$

and, with $y = d + x$,

$$\mathsf{MP}_x\langle t, w, d, h\rangle = \begin{cases} \langle 0, w, 0, y\rangle, & I(t, y) = h, \\ \langle 0, w, y, 0\rangle, & I(t, y) \neq h. \end{cases} \tag{Q5}$$

Only Shift can halt. The new axiom command, with $D = d + x$, $H = h + y$, $A = c$, has effect

$$\mathsf{AX}^{\mathrm{Q}}_{c,x,y}\langle t, w, d, h\rangle = \langle K(H, D, c), [q_1, \ldots, q_9, t; w], 0, 0\rangle,$$

where

$$(q_1, \ldots, q_9) = (\Gamma(A), L^{\mathrm{Q}}_6, L_5, L_4, L_3, L_2, L_1, \rho_8(D, A, H, d), U(d, h)). \tag{Q6}$$

Here $K(a, b, c) = G(a, b, c)$ is the unchanged guarded gate defined in the following all-axiom section, and $\Gamma(B) = I(B, U(B, B))$. The saved top remains the tenth pushed entry. All q_j except q_9 have residue zero modulo four; q_9 has residue two. The gate has residue zero, including a failed guard's output zero. Hence none of these generated entries is one.

The general cached-command proof depends on those residues, the ten positions, and the last two entries. It does not depend on the internal curried-premise order of q_2. For the new AX command define

$$\begin{aligned} \mathsf{Push} &= \mathsf{AX}^{\mathrm{Q}}_{0,0,0}; S^9; \mathsf{MP}_0, \\ \mathsf{Pop} &= \mathsf{AX}^{\mathrm{Q}}_{0,0,0}; S^{11}, \\ \mathsf{Discard} &= \mathsf{MP}_1; \mathsf{AX}^{\mathrm{Q}}_{1,0,0}; S^{10}; \mathsf{MP}_0. \end{aligned} \tag{Q7}$$

These words satisfy, for all displayed natural values,

$$\begin{aligned} \mathsf{Push}\langle t, w, d, h\rangle &= \langle 0, [t; w], 0, 0\rangle, \\ \mathsf{Pop}\langle t, [a; w], d, h\rangle &= \begin{cases} \mathsf{halt}, & t = 1, \\ \langle t, w, 0, a\rangle, & t \neq 1, \end{cases} \\ \mathsf{Discard}\langle t, w, 0, 0\rangle &= \langle 0, w, 0, 0\rangle. \end{aligned} \tag{Q8}$$

Indeed, the first ten Shifts after AX test the cleared cache and the nine generated entries, and never test the saved top. The next Shift tests precisely that top. Push's final MP necessarily fails because an implication has residue zero and the current universal cache has residue two. Discard's two failed comparisons follow from $I(t, 1) > 0$ and $I(U(1, 0), 0) > 0$. These calculations use the total projection identities, including their zero aliases; they do not require stacks to be free lists.

For $1 \leq j \leq 7$, the head selector remains

$$\mathsf{Discard}; \mathsf{AX}^{\mathrm{Q}}_{A,D,H}; S^{j+1}; \mathsf{Pop}^{9-j}. \tag{Q9}$$

It returns q_j on the original arbitrary stack with both caches zero, starting from any $\langle t, w, 0, 0\rangle$, even when $t = 1$. In particular the selector for the changed head is S^3 followed by seven Pops after Discard and AX. It returns the new head, not the old one; the latter is obtained by compiling the derivation in (Q4). The unchanged generalization, restricted-propositional-head, and gate selectors have the same proofs and exponents as PA194C.

For completeness, with Offset_0 empty and $\mathsf{Offset}_i = \mathsf{MP}_i$ for $i > 0$, generalization uses

$$\mathsf{Gen}_i = \mathsf{Push}; \mathsf{Offset}_i; S; \mathsf{AX}^{\mathrm{Q}}_{i,0,0}; S^{10}.$$

It sends $\langle a, w, 0, 0\rangle$ to $\langle U(i, a), w, 0, 0\rangle$ for every a, w, including $a = 1$. The MP suffix $S; S; \mathsf{MP}_b; S$ on

$$\langle 0, [a, I(a, b), 0; w], 0, 0\rangle$$

halts if $a = 1$ or $b = 1$, and otherwise returns $\langle b, w, 0, 0\rangle$. The dummy zero is essential for preserving the arbitrary underlying stack. The calculation also covers $a = b = w = 0$ and $I(0, 0) = 0$.

5.4 All-store completeness and syntactic soundness

For a finite $\mathcal{C}_{\mathrm{Q}}$ proof e, compile axiom leaves by their selectors, generalization by the preceding word, and an MP node with major proof e_1 of $I(a, b)$ and minor proof e_0 of a by

$$\mathsf{Discard}; \mathsf{Push}; W_{e_1}; \mathsf{Push}; W_{e_0}; \mathsf{Push}; S; S; \mathsf{MP}_b; S.$$

Induction on the finite proof tree proves the exact invariant: from every $\langle t, w, 0, 0\rangle$, the word W_e either halts or returns $\langle \mathrm{concl}(e), w, 0, 0\rangle$. Therefore, if e proves one, the word

$$\mathsf{AX}^{\mathrm{Q}}_{0,0,0}; W_e; \mathsf{Pop} \tag{Q10}$$

halts from every incoming store. This all-store assertion handles the inherited scheduler's retention of the store between unsuccessful trials. An assumption that every trial resets the store would be wrong. The unchanged exhaustive scheduler eventually tests the word (Q10), so a refutation yields a source halt.

Conversely annotate the visible top, cached head, and each live canonical stack entry by derivations in $\mathcal{C}_{\mathrm{Q}}$. Zero has the fixed derivation $\rho_8(0, 0, 0, 0) = 0$, so a single zero proof supplies the total stack's padding. AX emits its own axiom heads, the saved theorem, and a generalization of the cached theorem; successful MP applies modus ponens; failed MP returns zero. Shift moves derived formulas and halts only at a cached proof of one. This finite proof-store invariant extracts a refutation from every source halt. The changed head is sound in this invariant because it is an axiom of the new calculus, and by (Q4) it also has a proof in the old calculus.

These constructions are primitive recursive. Their correctness uses induction on finite proofs, total Cantor-stack decoding, natural loop counters, and finite traces. PA consequently proves

$$H_{R_{\mathrm{Q}}} \leftrightarrow \exists e\ \mathrm{Prf}_{\mathcal{C}_{\mathrm{Q}}}(e, 1), \qquad H_{R_{\mathrm{C}}} \leftrightarrow \exists e\ \mathrm{Prf}_{\mathcal{C}}(e, 1). \tag{Q11}$$

Here R_{Q} is the newly bound source program and R_{C} the pinned cached-head source. Equation (Q4), not source-step identity, connects their existential halting predicates.

5.5 Literal machines and the PA biconditional

The machine-specific certificates bind the literal PA194Q table to R_{Q}: every one of its 4,749 source outcomes has a positive-length, non-premature-halting simulation ending in the exact intended complete

frame, and its verified blank startup reaches the clean source seed. The corresponding interfaces for PA194C are retained from the pinned source packet. The generic arithmetic proofs of the literal transition and register-scan certificates yield

$$\mathrm{PA} \vdash H_{M_{\mathrm{Q}}} \leftrightarrow H_{R_{\mathrm{Q}}}, \qquad \mathrm{PA} \vdash H_{M_{\mathrm{C}}} \leftrightarrow H_{R_{\mathrm{C}}}. \tag{Q12}$$

The forward trace construction concatenates the physical simulations and prepends startup. For the reverse implication, a first literal halt at time T contradicts $T+1$ nonhalting source steps: their positive physical durations would give, by determinism, a nonhalting literal execution extending beyond T. This is a bounded arithmetic argument and does not assume either machine actually runs forever.

Combining (Q4), (Q11), and (Q12) proves the new preservation theorem

$$\boxed{\mathrm{PA} \vdash H_{M_{\mathrm{Q}}} \leftrightarrow H_{M_{\mathrm{C}}}.} \tag{Q13}$$

The supplied calibration of $\mathcal{C}$ gives primitive-recursive translations between its refutations and PA refutations, with PA-provable correctness. Its reverse translation uses arithmetic finite bit comprehension as a proof-producing induction template, not a uniform truth predicate. Its forward translation uses the diagonal-freshness replacement of equality vacuity and the established PA-refutation compiler. Using that inherited mathematical calibration with (Q13) gives

$$\boxed{\mathrm{PA} \vdash (\neg H_{M_{\mathrm{Q}}} \leftrightarrow \mathrm{Con(PA)}).} \tag{Q14}$$

This is an ordinary mathematical proof with executable finite and universal symbolic certificates. The new exchange and transport arguments do not turn the repository's conditional iPRA endpoints into completed formalizations. No newly compiled end-to-end Lean or Rocq theorem for (Q14) is asserted.

6 The inherited numerical theory and PA calibration

For reference, the unchanged constructors besides I, U are $M(a,b) = 4p(a,b)+1$, $E(a,b) = 4p(a,b)+3$, and $N(a) = I(a,1)$. Define $a \wedge b = N(I(a, N(b)))$ and $\exists_i B = N(U(i, N(B)))$. The six ordinary heads of the new theory are

$$\begin{aligned}
L_1 &= I(\exists_D U(D,H), H),\\
L_2 &= I(U(D, U(A,H)), U(A, U(D,H))),\\
L_3 &= I(M(H,D), I(E(H,A), M(A,D))),\\
L_4 &= I(E(H,D), I(M(A,H), M(A,D))),\\
L_5 &= \exists_D E(H,D),\\
L_6^{\mathrm{Q}} &= I(U(D, I(A,H)), I(U(D,A), U(D,H))).
\end{aligned}$$

The remaining implicational schema is

$$\rho_8(a,b,c,u) = I(I(I(c,u),a), I(I(a,c), I(b,c))), \qquad u \le a,$$

and the separate diagonal head is $\Gamma(B) = I(B, U(B,B))$. The guarded separation head uses

$$\begin{aligned}
V(a,b,c) &= I(M(a,b), U(c, M(a,b))),\\
P(b,c) &= I(c, M(c,b)),\\
S(a,b,c) &= U(b,\ I(M(a,b), P(b,c)) \wedge I(M(c,b), c)),\\
Q(a,b,c) &= U(c, I(S(a,b,c), N(P(b,c)))),\\
G(a,b,c) &= \begin{cases} 0, & c = a \text{ or } c = b,\\ N(I(V(a,b,c), Q(a,b,c))), & \text{otherwise.} \end{cases}
\end{aligned}$$

Parameters range over natural codes. The rules are numerical modus ponens and generalization $B \mapsto U(i, B)$. These are the inherited schemas with only L_6 changed. The command-emitted $U(d, h)$ is justified by generalization of the already derived cached head; it is not an unrestricted axiom asserting every universal formula.

The supplied calibration translates finite refutations between this separate-head numerical theory and PA. On the arithmetic side, membership is interpreted by binary-bit membership and primitive equality by ordinary equality. PA proves finite bit comprehension for each formula using an induction instance. Replacing the formula in that finite derivation is a proof-producing syntactic operation, rather than a uniform truth predicate. The collision $a = b$ requires its separate singleton-comprehension argument. On the reverse logical side, formula-code freshness justifies the diagonal clause: every label occurring in a formula of code B is less than B, with the zero-code case handled separately. The supplied proof removes equality from a quantified consequent before using membership-only vacuity, so it does not assume the equality-vacuity principle it is trying to derive.

The exact retained manuscripts are in `references/calibration/` and `references/cached_commands.tex`. They are mathematical dependencies, not newly verified formal theorems in this packet. The present proof adds an explicit h6 proof translation and a new literal implementation, leaving those dependencies visible and reviewable.

7 A second construction: symmetric Shift with a zero-register jump

The optional second machine, PA194CSZ, combines the supplied symmetric-Shift controller PA194CS with the supplied zero-register realization of one source jump. It has 194 working controls and 388 transitions. The construction is not a new sub-194 result. Its contribution is the new composition, a complete common invariant certificate, and a fresh independent literal-tape audit.

7.1 Exact source and physical changes

Use the logical register order

$$(r_0, \ldots, r_8) = (N, T, W, A, D, H, X, Y, L).$$

The supplied PA194CS source has 3,646 primitive sites. In its source tree, `node22` has the two leaves `dec 8` and `breakTo 1`. Replace the latter leaf by `dec 6`. The entire resolved source map changes at just site 26:

PA194CS	PA194CSZ
$26 : \text{jump}\, 34$	$26 : \text{dec}\, r_6\, [34, 27].$

The first decrement successor is its zero branch. Every other opcode, register, and successor is identical. The new source still has 3,646 sites and now has 4,746 primitive branch outcomes, one more than its parent.

The literal native tables differ in just one row:

$$(27, 1): \quad 0L40 \quad \longmapsto \quad 1R21.$$

Native start and halt are 74 and 254. The delivered dense numbering uses start 0, working states $0, \ldots, 193$, and separate halt label 194. The head begins at cell zero on a two-way all-zero binary tape.

7.2 A finite common invariant proves source preservation

For a concrete register vector $r \in \mathbb{N}^9$, define its positivity mask

$$\mu(r) = \sum_{i=0}^{8} 2^i [\, r_i > 0 \,] \in \{0, \ldots, 511\}.$$

The finite abstract transition system acts on pairs (p, m) of a source site and a positivity mask. A jump retains the mask. Incrementing r_i sets bit i. A decrement with bit i zero takes the zero successor and retains the mask; with bit i one it takes the positive successor, with bit i either retained or cleared. A halt has no successors. Every concrete step projects to one of these abstract steps. In the positive decrement case, the two abstract possibilities distinguish $r_i = 1$ from $r_i > 1$; no bounded-register assumption is involved.

Starting with $(25, 1)$, the two source graphs have the same abstract reachable closure R, containing exactly **263,180 pairs**. The delivered finite certificate checks all of the following:

1. $(25, 1) \in R$;
2. R is closed under every abstract successor for both graphs;
3. the two abstract successor sets agree at every element of R;
4. the 32 masks occurring in R at site 26 all have bit 6 zero.

The first checker computes the two closures independently. The second checker reads the finite relation and verifies its seed, range, closure, common successors, and site-26 property without importing the builder.

The concrete clean source seed is

$$(p, r) = (25, (2, 0, 0, 0, 0, 0, 0, 0, 0)),$$

whose mask is 1. Induction on execution length and abstract-step soundness therefore prove $r_6 = 0$ whenever either reachable execution visits site 26. At that site, the new decrement consequently leaves the complete register vector unchanged and goes to site 34, exactly like the old jump. All other source instructions are equal. A second induction shows that the two concrete source executions have identical program counters and register vectors at every finite step, including the same halting outcome.

This proof makes no consistency assumption and uses no empirical observation of a long blank run. The masks overapproximate unbounded register values. The relation is finite and explicit, and the projection and induction are primitive recursive arithmetic reasoning available in PA.

7.3 Physical realization and exact initialization

The fully unfolded table has 202 controls: 135 fixed controls and 67 AX readers. The seven original decoder folds and the symmetric-Shift fold $41 \mapsto 330$ are applied together, leaving

$$135 + 67 - 8 = 194.$$

All 32 two-cell local cases are verified against the final simultaneous quotient, with arbitrary unchanged exterior tape; the maximum detour has five physical transitions. Thus the new jump realization has not invalidated an earlier fold by changing a helper action.

The original symbolic runtime verifies all 4,746 primitive outcomes for both the raw and folded tables. For the folded table it uses 1,513,882 symbolic transitions, with at most 508 per outcome. The extra

positive branch at site 26 is checked even though the invariant excludes it in the blank-start source execution.

A separate literal-tape checker rederives four generic list-sweep lemmas from the actual transition rows and checks all source outcomes without the first checker's arithmetic-block oracle. Its complete replay uses 1,100,566 symbolic transitions, at most 403 per outcome, with minimum positive-duration constant 3. It also resolves the source graph in two independent ways. The contracts quantify over nine arbitrary natural register values, header distance $48 + n$ with $n \geq 0$ (zero gap $48 + n - |p|$ for address p, as in the principal physical-frame convention), and an arbitrary nonempty finite spare list. They are not tests over bounded numerical examples.

The folded machine reaches the complete clean source frame after exactly **58,014** physical transitions. The independent bytearray simulator compares the entire tape, not merely the main registers or a window. The head is at -2 in native state 84, the source site is 25, the logical register vector is the seed above, and all 26 spare registers have value one. No earlier halt occurs. The fully unfolded table reaches its clean seed after 57,840 transitions. The old symmetric-Shift table reaches its seed after 57,768, so no speed improvement is claimed for this composition.

Concatenating positive physical source-step simulations, and prepending initialization, proves that each literal table halts precisely when its source program halts. Conversely, a literal halt after t transitions cannot coexist with $t + 1$ nonhalting source steps: their positive simulations would deterministically reproduce more than t nonhalting physical steps. Together with the common-source invariant, this gives the new transfer

$$\mathrm{PA} \vdash \mathrm{Halt}(\mathrm{PA194CSZ}) \longleftrightarrow \mathrm{Halt}(\mathrm{PA194CS}).$$

The inherited ordinary mathematical calibration of PA194CS then transfers its nonhalting/consistency biconditional to PA194CSZ. This does not claim a new end-to-end Lean or Rocq kernel theorem, or completion of the repository's conditional iPRA endpoint.

7.4 Identity and provenance

The authoritative dense-table SHA-256 is

```
62e93098caf7c09ff2938d07064f0dbbb47c71ec369892968258fff708592269.
```

Its action graph is absent from the frozen current corpus of 620 literal byte forms, including all eight old 194-state graph classes. The comparison allows arbitrary control renaming, a changed designated start, global tape reflection, and simultaneous read/write symbol complementation. This finite novelty check is independent of the semantic preservation argument.

Symmetric Shift and its additional fold come from `pa194_new_designs_2026-11-13.zip`. The zero-register jump and positivity-mask method come from `pa194c_conpa_2026-11-13 (1).zip`. Their authorship and supplied licenses remain attached. The new construction proves their compatibility, checks the resulting literal machine, and supplies the independent audit that the earlier auxiliary zero-register seed did not include.

8 Search below 194 and limits of the negative result

The discovery work attacked both parts of the state budget. Source-level searches changed safe axiom presentations, including premise exchange, reversal of primitive relation coding, and equivalent separation matrices. Decoder searches reassociated the emitter while retaining valid address and

return-scanner constraints. Further searches expanded formerly opaque copy macros, reused fixed reader subtrees, counted only graph-reachable controls, and varied independent parameter roles within head families.

The local-state searches used complete finite windows from source simulations, including arithmetic-block traces, and considered common-action state identification as well as eliminating a state by replacing each incoming transition with a bounded path reproducing its two-step effect. Their failure is limited to the specified local sufficient rules. It does not rule out a different encoding, a stronger invariant, a longer local simulation or a different proof-search program.

Completed root search	Proposal slots	Best reader objective
Axiom-head variants	2,832,997	60
Focused axiom-head search	1,664,002	65
Separation-matrix variants	1,761,744	61
Head-parameter permutations	1,766,625	60

The reader objective is the number of raw decoder readers minus a simultaneously checked fold set; the fixed part adds 134 controls. Proposal slots are neither distinct machines nor proofs. Some proposed rotations are skipped. The accepted PA194Q endpoint has objective 60 and separately passes all source and literal checks. The search logs and random seeds are retained.

The expanded-macro branch additionally records 15,895,250 proposal slots in completed runs and at least 4,572,000 before interrupted runs. Its strongest local screen first proves a finite ternary register invariant (0, 1, or at least 2): 3,648 sites and 4,747 primitive outcomes are admissible from the exact clean seed. Specialized independent literal replay checks 4,749 affine cases (two require a zero/one split), covering all 194 controls and yielding 29,396 conservative radius-five windows. Common-action identification, per-predecessor two-step state deletion, and rejoining after 2–24 original transitions all produce empty witness lists under their stated finite restrictions. The source invariant, complete bootstrap binding, windows and portable enumerators are in `search/invariant/`. None is a global state lower bound.

Several tempting logical reductions were deliberately excluded. Replacing the equality-witness head by reflexivity loses the ability to transfer between variable domains; the supplied unequal-domain countermodel satisfies the retained schemas but refutes the witness. Replacing the first head $\Diamond\Box H \to H$ by universal elimination $\Box H \to H$ plus explosion requires a new argument: the former expresses modal symmetry, which does not follow from reflexivity in normal modal logic. The existing calibration uses that symmetry to extend vacuity to negations. A syntactically shorter emitter without the missing proof would not be a consistency construction.

The ZFC work supplies useful history-based decoder folds, physical-frame verification and register-interface reuse. Its logical calibration cannot be substituted for PA's. Likewise a small iPRA proof checker searches for iPRA inconsistency, not PA inconsistency. One longer-term route is to extend the concise intuitionistic system's equation-restricted progress rule to arbitrary formulas, thereby supplying full HA induction, and then use a syntactic negative translation between PA and HA refutations. This would require its own precise theory, proof compiler and binary machine; no small-state result is claimed for that route here.

9 Further research questions

1. Can the new sixth-head emitter be represented by 66 raw readers with seven compatible folds, or 67 with eight, while retaining its current return scanner? Either target would give 193 states.

2. Can a proved source invariant eliminate a runtime control rather than only a source instruction? The final guard has tighter reachable value bounds than its arbitrary-register contract. An optimization must bind the invariant to the exact source and transport it through literal frames.

3. Can macros be expanded selectively so that reader reuse saves a fixed control as well as sharing new decoder subtrees? Full expansion can destroy useful sharing; a mixed representation deserves a cost model counting all reachable controls and compatible folds.

4. Can the six axiom heads and the guarded separation matrix be optimized jointly with their proof translations? The new implementation demonstrates that a logically equivalent presentation can change physical sharing even at identical source length. A useful search should retain a proof witness for every permitted formula rewrite.

5. Can the physical contracts, formula-code calibration and all-store command compiler be connected in one Lean or Rocq development, with an explicit PA or iPRA derivation as the endpoint? The existing conditional iPRA results and finite-run theorems are reusable components, but their missing machine/refutation bridges must still be supplied.

10 Reproduction, files and attribution

Python 3.10 or later and its standard library suffice for the accepted construction checks. Run without `-O` or `PYTHONOPTIMIZE`, since assertions are part of the certificate checks. From the archive root:

```
python3 -B reproduce.py
python3 -B reproduce.py --secondary
```

The first command deterministically rebuilds PA194Q, replays the 61-line exchange certificate, checks the exact source/formula binding, repeats the literal physical audit and checks graph novelty against the frozen corpus. The second additionally rebuilds and verifies PA194CSZ and its invariant. The independent literal audits can take several minutes. They need no network, external repository, solver or installed proof assistant.

Location	Contents
`machine/`	Complete dense and native PA194Q tables, raw parent, labels, source map and word.
`construction/`	Frozen grammar, folds, deterministic compiler and credited runtime references.
`verification/`	Independent literal checker, logical block audit and symbolic exchange certificate.
`evidence/`	Exact construction, source, local-fold and whole-tape receipts.
`comparison/`	Full fresh corpus census, 41-packet ledger, old frontier tables and novelty checker.
`secondary/pa194csz/`	Complete optional second construction, independent invariant checker and replay.
`references/`	Exact inherited mathematical calibration and implication-basis references.
`search/`	Discovery inputs, C++ sources, bounded logs and source-variant notes.
`report/`	This PDF, main LaTeX source and every included fragment.

The construction builds on James Chang's PA machines and the supplied ProveIt research packets, including their compiler, runtime, symbolic engines and logical calibration. Original license and provenance files are retained under `licenses/` and alongside inherited components. The new work consists of the concrete PA194Q realization, source/proof binding, audits and comparison, and the independently verified CSZ composition. No repository changes have been published.

A Complete PA194Q transition table

An action wLq or wRq writes w, moves left or right, and enters q. H denotes halt label 194. The initial state is 0. Four blocks are printed side by side; together they contain all 194 working states.

State	Read 0	Read 1	State	Read 0	Read 1	State	Read 0	Read 1	State	Read 0	Read 1
0	1R1	1R2	49	0R68	0R69	98	0R111	1R75	147	0R155	1R146
1	1R3	1R1	50	0R70	0L48	99	0L63	0R109	148	0R156	1R157
2	0R0	1R2	51	0R64	1R53	100	0L62	1R113	149	0R158	1R159
3	0R4	0R1	52	0L71	0L15	101	0L90	1R114	150	0L159	1R160
4	1R5	1L6	53	0R72	0L10	102	0R115	0L14	151	0R41	1L161
5	1L7	1L8	54	0R73	1R74	103	0R69	1R103	152	0R162	1R163
6	0L7	1L6	55	0R75	1R76	104	0R116	1R95	153	0R164	1R164
7	0L9	1L6	56	0R77	1R78	105	0R116	1R67	154	0R165	1R166
8	0L10	0L11	57	0R79	1R80	106	0R117	0L14	155	0R92	1R111
9	1R12	0L9	58	0L81	0R1	107	0R118	1L71	156	0R76	1R165
10	0L9	0L10	59	0R82	1R59	108	0R119	1R120	157	0R167	1R165
11	1L8	1L11	60	0L73	1L46	109	0R14	1R109	158	0R168	1R169
12	0R12	0R13	61	0R83	1R84	110	0R66	1R110	159	0R170	1R157
13	0L14	0R13	62	0R54	1R85	111	0R121	1L35	160	0R135	1R130
14	1L15	0L15	63	0L86	1R87	112	0R122	1R123	161	0R138	1R135
15	0L15	1L16	64	0R58	0R82	113	0R88	0L14	162	0R171	1R172
16	0L17	1L16	65	0R88	1R30	114	0R90	1L95	163	0R173	1R172
17	0L18	1L16	66	0R89	1R66	115	0R124	1R124	164	0L174	1R175
18	0L19	1L16	67	0L90	1L33	116	0L63	0R103	165	0L162	1L176
19	0L20	1L16	68	0L81	0R2	117	0R68	0R91	166	0R161	1R176
20	0L21	1L16	69	0R91	1R69	118	0R58	0R91	167	0R97	1R177
21	0L22	1L16	70	0R72	1R92	119	0L86	1R125	168	0R134	1R178
22	0L23	1L16	71	0R46	1R93	120	0R126	1R127	169	0R150	1R163
23	0R23	1R24	72	0R42	0L13	121	0R58	0R110	170	0R179	1R180
24	0R25	1R26	73	0R94	1R37	122	0R99	1R113	171	0R167	1R140
25	0R27	1R28	74	0R30	1R95	123	0R79	1R78	172	0R134	1R145
26	0R29	1R30	75	0R42	1R96	124	0R88	1R88	173	0R181	1R41
27	0R31	1RH	76	0R97	1R98	125	0R128	1R129	174	0R175	1R182
28	0L32	1R33	77	0R99	1L100	126	0R32	1R36	175	0R167	1R145
29	0R34	1R35	78	0R30	1R77	127	0R31	1R111	176	0R183	1R173
30	0L36	1L37	79	0R93	1R101	128	0R130	1R131	177	0R107	1R85
31	0R38	1R39	80	0R30	1R102	129	0R132	1R133	178	0R184	1R165
32	0R40	1R41	81	1R81	1L15	130	0R134	1R135	179	0R134	1R181
33	0R42	1R43	82	0R103	1R82	131	0R136	1R137	180	0R185	1R185
34	0R44	1R45	83	0R94	0L7	132	0R138	1R107	181	0R167	1R151
35	0R46	1R33	84	0R104	1R105	133	0R139	0L10	182	0R151	1R130
36	0R47	1R48	85	0R42	1R106	134	0R97	1R35	183	0R164	1R186
37	0R49	1R50	86	0R107	1R48	135	0R140	1R124	184	0R140	1R187
38	0R51	1R42	87	0R108	0L72	136	0R141	1R142	185	0R188	1L189
39	0R52	1R53	88	0R68	0R66	137	0R143	1R144	186	0R190	1R41
40	0R54	1R55	89	0R109	1R89	138	0R145	1R41	187	0R32	1R189
41	0R56	1R57	90	0R49	1R100	139	0R146	1R147	188	0R88	1R191
42	0R58	0R59	91	0R110	1R91	140	0L148	1L143	189	0R148	1R192
43	0R60	0L14	92	0R30	1R48	141	0R28	1R149	190	0R41	1R138
44	0R61	1R62	93	0R47	1R60	142	0R135	1R150	191	0R174	1R88
45	0R63	1R52	94	0L63	0R89	143	0R151	1R152	192	0R182	1R193
46	0R64	1R65	95	0R94	1L111	144	0R153	1R154	193	0R161	1R138
47	0R58	0R66	96	0R101	0L14	145	0L130	1R88			
48	0R67	0L14	97	0R36	1R112	146	0R31	1R52			

B Complete PA194CSZ transition table

The conventions are identical to Appendix A.

State	Read 0	Read 1	State	Read 0	Read 1	State	Read 0	Read 1	State	Read 0	Read 1
0	1R1	1R2	49	0L66	0L15	98	0R112	1R104	147	0R149	1R158
1	1R3	1R1	50	0R67	0L10	99	0R113	0L14	148	0R159	1R46
2	0R0	1R2	51	0R68	1R69	100	0R114	1L66	149	0R91	1R160
3	0R4	0R1	52	0R70	1R71	101	0R115	1R116	150	0R161	1R162
4	1R5	1L6	53	0R72	1R73	102	0R117	1R102	151	0R163	0L10
5	1L7	1L8	54	0R74	1R75	103	0R14	1R103	152	0R164	1R165
6	0L7	1L6	55	0L76	0R1	104	0L109	1L33	153	0R166	1R157
7	0L9	1L6	56	0R61	1R56	105	0R67	1R118	154	0R167	1R156
8	0L10	0L11	57	0L68	1L35	106	0R119	1L120	155	0R135	1R130
9	1R12	0L9	58	0R77	1R78	107	0R121	1R122	156	0R147	1R40
10	0L9	0L10	59	0R51	1R79	108	0R46	0L14	157	0R134	1R168
11	1L8	1L11	60	0L80	1R81	109	0R85	1R94	158	0R40	1L169
12	0R12	0R13	61	0R82	1R61	110	0R109	1L89	159	0L170	1R171
13	0L14	0R13	62	0L76	0R2	111	0R123	1R123	160	0R100	1R79
14	1L15	0L15	63	0R83	1R63	112	0L60	0R82	161	0R172	1R170
15	0L15	1L16	64	0R30	1R84	113	0R62	0R117	162	0R173	1R174
16	0L17	1L16	65	0R85	1R86	114	0R55	0R117	163	0R175	1R176
17	0L18	1L16	66	0R35	1R87	115	0L80	1R124	164	0R177	1R178
18	0L19	1L16	67	0R41	0L13	116	0R125	1R126	165	0L179	1L180
19	0L20	1L16	68	0R88	1R65	117	0R127	1R117	166	0R149	1R140
20	0L21	1L16	69	0R47	1R89	118	0R47	1R84	167	0R165	1R181
21	0L22	1L16	70	0R41	1R90	119	0R55	0R127	168	0L130	1R46
22	0L23	1L16	71	0R91	1R92	120	0R35	1R33	169	0R182	1R135
23	0R23	1R24	72	0R93	1L94	121	0R93	1R108	170	0R183	1R145
24	0R25	1R26	73	0R47	1R72	122	0R74	1R73	171	0R158	1R130
25	0R27	1R28	74	0R87	1R95	123	0R46	1R46	172	0R139	1R165
26	0R29	1R30	75	0R47	1R96	124	0R128	1R129	173	0R158	1R184
27	0R31	1RH	76	1R76	1L15	125	0R32	1R64	174	0R185	1R100
28	0L32	1R33	77	0R88	1R41	126	0R31	1R106	175	0R31	1R49
29	0R34	1R35	78	0R97	1R98	127	0R63	1R127	176	0R186	1R175
30	0L33	1L36	79	0R41	1R99	128	0R130	1R131	177	0R134	1R142
31	0R37	1R38	80	0R100	1R84	129	0R132	1R133	178	0R32	1R187
32	0R39	1R40	81	0R101	0L67	130	0R134	1R135	179	0R144	1R188
33	0R41	1R42	82	0R102	1R82	131	0R136	1R137	180	0R189	1R190
34	0R43	1R44	83	0R103	1R83	132	0R138	1R139	181	0R40	1R191
35	0R45	1R36	84	0R104	0L14	133	0R140	1R141	182	0R168	1R40
36	0R46	1R47	85	0R62	0R102	134	0R91	1R120	183	0R192	1R46
37	0R48	1R41	86	0R105	0L84	135	0R142	1R123	184	0R153	1R146
38	0R49	1R50	87	0R30	1R57	136	0R143	1R144	185	0R165	1R180
39	0R51	1R52	88	0L60	0R83	137	0R145	1R146	186	0R118	1R106
40	0R53	1R54	89	0R88	1L106	138	0R134	1R147	187	0R193	1R159
41	0R55	0R56	90	0R95	0L14	139	0R46	1R148	188	0R169	1R154
42	0R57	0L14	91	0R64	1R107	140	0R149	1L131	189	0R158	1R192
43	0R58	1R59	92	0R106	1R70	141	0R150	1R151	190	0R179	1R182
44	0R60	1R49	93	0L60	0R103	142	0R71	1R40	191	0R182	1R40
45	0R55	0R61	94	0L59	1R108	143	0R28	1R152	192	0R149	1R168
46	0R62	0R63	95	0L109	1R110	144	0L153	1L154	193	0R71	1R144
47	0L64	1L65	96	0R111	0L14	145	0L129	1R155			
48	0R45	1R50	97	0R112	1R89	146	0R156	1R157			

C Archive-by-archive catalogue

The following ledger covers all 41 delivered packets represented by the pinned current tree and incoming archives. Descriptions report the packets' contributions and evidence scope, rather than independently endorsing every claim or exploratory table. Exact origins, archive/member identities and hashes are retained in the machine-readable comparison files.

Packet	Contribution and reported status
`Below_U208_Research_2026-10-29.zip`	Five-site source pruning 3909 to 3904 and wider local-state searches; no accepted below 208 table. Prior U 208 saving already shared.
`ConPA_below208_research_2026-10-29.zip`	Two-object obstruction to collapsing cover roles; 279936 restricted parameter assignments and fold searches; no accepted below 208 table.
`PA195_ConPA_research_2026-11-14.zip`	195S uses 134+67-6 with original commands,3673 sites,4780 contracts,58154 startup. 195C uses same reader family with Combined commands,3649 sites,4749 contracts,57922 startup. Two mathematical constructions.
`PA_consistency_TM_research_2026-11-13.zip`	Five-zero source geometry and combined-negation/MP 157-node source; parameter-free MP compiler; no smaller physical allocation established.
`U208_ConPA_2026-10-29.zip`	Native 172/211 sharing produces 208 with unchanged 3909-site program and 48140 startup; main and compatibility variants.
`U208_ConPA_Conditional_Context_2026-10-29.zip`	Same original two U 208 graph classes and same single-state saving, expressed as conditional entry contexts.
`U208_ConPA_Entry_Context_2026-10-29.zip`	Same original U 208 reduction with positive local blocks of lengths 1,3,7; complete physical source evidence; savings not additive.
`U208_ConPA_History_2026-10-29.zip`	Same two U 208 classes with four exceptional blocks, unchanged source and initialization; failed additional reduction retained.
`conpa_tm_comparison_and_reduction_search.zip`	Two MP command 196 experiments; two failed Shift layouts actually halt on blank after 43563 steps. Different proof status must be retained.
`conpa_tm_comparison_and_search.zip`	Accepted A/D physical register-swap 196; two-rotation 195 violates 93 primitive contracts.
`conpa_tm_study_2026-11-13.zip`	H-only Shift plus MP-result-in-H gives 196,3788 sites,4928 contracts,57922 startup; direct-negation 197 and failed physical source-compression directions.
`pa194_conpa_2026-11-13(2).zip`	202-control parent with 134+68-8=194,3673 sites,4780 contracts,58154 startup; complete 32-case simultaneous quotient,7-step detour. Also distinct intermediate separate 195.
`pa194_conpa_2026-11-13.zip`	201-control parent with 134+67-7=194; same 3673-site source,4780 contracts,58154 startup. Secondary 194 swaps physical A/D. Complete 28-case simultaneous quotient.
`pa195_conpa_2026-11-13.zip`	202-control parent with 134+68-7=195;3700 sites,4815 contracts,58154 startup; full written PA calibration and physical certificates.
`pa195_diagonal_combined_2026-11-13.zip`	Six physical row edits produce PA 195DC,3676 sites,4784 contracts,57922 startup. Optional positive-rho permutation 195R and negative-vacuity 195N have separate logical scope; no below 195 result.
`pa195_followup_research.zip`	Eight-fold nominal 194 fails because address seven-zero run is confused with separator; extra scanner repairs to 135+68-8=195,3700 sites,4815 contracts,59164 startup.

Packet	Contribution and reported status
pa196_research_2026-11-13.zip	Original 3812-site CFG represented in 157 nodes, but missing contextual helper defeats prospective 195 allocation; no new smaller machine.
pa208_research_2026-10-29.zip	Eighteen further helper-layout graphs;27 designated presentations represent 20 U 208 graph classes. All use inherited consistency calibration.
pa_below196_research.zip	Eight equivalent gate spellings, membership/equality orientations, freshness obstruction and source/physical searches; no smaller validated table.
pa_consistency_comparison_2026_11_13.zip	Modal/explosion split with finite Hilbert equivalence certificates, twelve semantic compositions and countermodels to head deletions; no smaller implementation.
pa_consistency_comparison_and_new196_2026-11-13.zip	Diagonal 191 emitter:3700 sites; separate 190:3673; scanner-best 196 plus closing variants. Logical/source ancestors of positive 195 and 194 tables.
pa_consistency_cross_archive.zip	Four 196 combinations: diagonal/separate AX with Combined commands, optionally zero-jump;3676/3649 sites and 57922/58168 startup. Ten-entry selectors depend on command choice.
pa_consistency_reduction.zip	Rebracketed 196, H-only Shift 196 and larger integrated gate 203/206; two apparent 195 allocations rejected by universal runtime checks.
pa_consistency_reduction_audit_2026-11-14.zip	Further MP-in-H combinations and larger bounded logical/physical searches; no supported state reduction; experimental status retained.
pa_consistency_research.zip	Reassociation saves source-reader cost, but extra scanner required; valid 196. Nominal 195 returns wrong source frame.
pa_consistency_research_2026-10-29.zip	3909 to 3904 source pruning under proved zero guard; source compression does not reduce physical table below 208.
pa_consistency_research_2026-11-13(2).zip	New rotated 196, diagonal 198, successful-MP halting-command source experiment; latter has no newly compiled binary table.
pa_consistency_research_2026-11-13.zip	A to forall_A A gate reduces 201 to 191 AX macro items and 3812 to 3700 sites; first physical realization 199, also short-negation 197.
pa_consistency_research_2026-11-14.zip	18-source-instruction negation reduces separate source 3673 to 3441; full implementation 197,4468 contracts,58154 startup; source saving loses physical sharing.
pa_tm_comparison_and_reductions.zip	Two distinct rotated 196 layouts; gate 199-token source 3773 compiles to 198. Exact algebraic transformations and bounded search evidence.
pa_tm_extended_research_2026-11-13.zip	Four one-row closing variants; stronger register-zero/difference invariants. Variants change same row, so savings cannot be added.
pa_tm_research_2026-11-13(2).zip	Zerojump 196 changes source 26 branch under R 6=0,4959 contracts,58402 startup. Deleting extra helper yields wrong register/successor, even on blank execution.
pa_tm_research_2026-11-13.zip	Alternative 196 keeps 3812-site source but lowers maximum path depth 38 to 36; startup/runtime checks and written reverse-PA translation; pruned 198 lacks full startup calibration.
pa_unary_power_cover.zip	One-free-variable matrix theory with uniform finite-proof translations to/from PA; new numerical theorem stream, no new literal compiled machine.

Packet	Contribution and reported status
u208_comparison_research_2026-10-29.zip	Original nine designated tables give two rooted classes; attempted 207 fails a legal physical frame. No accepted below 208.
u208_research_2026-10-29.zip	Same original history fold, primary/alternative U 208,5084 symbolic contracts and 48140 startup; no additional independent state saving.
pa194C_conpa_2026-11-13(2).zip	Duplicate principal C table d 759313c...; complete symbolic/independent physical checks, mathematical PA proof claim.
pa194_new_designs_2026-11-13.zip	C repeats existing principal graph. CS removes 3 sites, then new local 41 to 330 fold recovers additional reader; principal graph 1c 71002e... .
pa194c_conpa.zip	Principal C is identical. Includes distinct pa 194altC source-preserving search input with full original symbolic and seed audit, without independent lemma rederivation.
pa194c_conpa_2026-11-13(1).zip	Principal C identical. CZ adds native 27/read 1 to 1R 21 guarded by register 6=0 at source 26; invariant and original physical audit, no separate independent replay.
pa194c_conpa_2026-11-13.zip	Principal C identical; includes failed scanner-removal experiments and bounded grammar search.

References

[1] ProveIt, PA consistency machines, README at the pinned commit, https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/Computability/BusyBeaver/Research/PA.

[2] ProveIt incoming research packets at the pinned commit, https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/docs/incoming.

[3] James Chang, imported PA196 implication basis, PA196/Logic/ImplicationBasis.lean, in the pinned repository's Computability/BusyBeaver/Research/PA/jamesyc-upstream/pa196/. The exact relevant source is retained in references/.

[4] Supplied PA195 diagonal-combined packet, pa195_diagonal_combined_2026-11-13, search/source_variants.py and report/sections/logical_experiments.tex; see also the supplied reduction-audit packet's logical-variant search. Premise exchange is prior work; the certified 194-state realization here is new to the pinned corpus.

[5] Supplied pa194_new_designs_2026-11-13.zip, PA194C and PA194CS construction, complete cached-command and logical calibration manuscripts. Relevant exact dependencies are retained in references/ and the secondary package.

[6] ProveIt ZFC machine research, https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/Computability/BusyBeaver/Research/ZFC.

[7] ProveIt ConciseIPRA, https://github.com/VladimirReshetnikov/ProveIt/tree/17f92105e2bc4317feae6c219cb2143437b482ff/Logic/PrimitiveRecursiveArithmetic/ConciseIPRA.